

MODEL RESEARCH DOCUMENT

Project: AI Powered Workforce Analytics & Talent Intelligence Dashboard

The project aims to develop an AI-Powered Workforce Analytics & Talent Intelligence Dashboard to provide organizations with actionable insights into key areas like workforce performance, employee engagement, and talent development. By leveraging advanced technologies such as Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and predictive analytics, the solution transforms fragmented HR data into strategic business intelligence. The platform is designed to automate real-time analytics and support conversational querying, allowing HR leaders and executives to monitor metrics like diversity, attrition, and recruitment effectiveness. Ultimately, this tool empowers decision-makers to proactively address talent challenges, optimize workforce planning, and refine retention strategies based on data-driven evidence. By integrating data from various HR systems and providing intelligent recommendations, the project focuses on enhancing overall organizational health and operational performance.

Data source : <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

Model	Primary Use Case in Your Dashboard	Typical Accuracy Range
Logistic Regression	Baseline performance; provides high interpretability for explaining to HR leaders which specific factors drive attrition.	80% - 85% Accuracy
Random Forest	Handles complex, non-linear relationships between employee attributes; very robust against overfitting in HR datasets.	85% - 89% Accuracy
XG Boost	High-performance boosting model; best for maximizing predictive accuracy when dealing with intricate patterns in employee data.	87% - 92% Accuracy
Gradient Boosting	Similar to XGBoost, it focuses on reducing bias by correcting errors sequentially, which helps improve predictions for minority classes	86% - 90% Accuracy
Support Vector Machine(SVM)	Effective for classification in high-dimensional feature spaces, helpful if you include many unique employee attributes or categorical variables.	83% - 87% Accuracy
Decision Tree	Offers a visual "tree" structure that acts as a simple decision rule, making it useful for explaining specific attrition "paths" directly to non-technical stakeholders.	78% - 83% Accuracy

Important Implementation Notes for Our Dashboard:

- **Accuracy:** These percentages are typical ranges for this specific dataset but will vary based on how you perform feature engineering (e.g., encoding, scaling) and how you handle the class imbalance—attrition is often the "minority" class, so you may need techniques like SMOTE (Synthetic Minority Over-sampling Technique) or class weighting to improve performance.
- **Interpretability vs. Performance:** While models like **XGBoost** provide the highest accuracy, **Logistic Regression** or **Decision Trees** are significantly more valuable for "explainability," which is critical when HR managers need to justify retention interventions based on model insights.

Conclusion: Strategic Model Selection for Workforce Intelligence

- **Balancing Performance & Explainability:** The project utilizes a tiered model strategy, prioritizing high-accuracy algorithms like **XGBoost** for predictive precision, while leveraging **Logistic Regression** and **Decision Trees** to provide the transparent, explainable insights essential for HR stakeholder adoption.
- **Validated Foundation:** By benchmarking models against the **IBM HR Analytics dataset**, we establish a rigorous baseline for predicting attrition and organizational health, ensuring the engine performs reliably with real-world employee data.
- **Proactive Retention Framework:** These predictive models transition HR from reactive report generation to proactive talent management, allowing organizations to identify at-risk talent and implement data-backed interventions early.
- **Actionable Intelligence:** Integrating these models into the dashboard's conversational interface empowers leaders to query workforce risks in natural language, reducing the time from insight to action.
- **Scalable Architecture:** The selected models are designed for continuous refinement, utilizing automated feedback loops to improve prediction accuracy as the organization's workforce dynamics evolve.